

HCSM-36

The PMNS Matrix

The Spin Structure Construction, the Neutrino $N/4$ Spectrum (60, 60, 58), and the Generation Phase from the Self Dual Torus

Hodge Complex Standard Model

Stanley Preschutti

Entropy Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

September 2026

Abstract

We derive the Pontecorvo-Maki-Nakagawa-Sakata (PMNS) matrix from the geometry of the 8×8 torus using the same framework constants and the same mass matrix construction that produces the CKM matrix (HCSM-34). The lepton sector requires one new ingredient not present in the quark sector: the three neutrino $N/4$ labels. We derive them from the spin-structure construction of the torus.

The torus T^2 admits four spin structures with Dirac indices $\pm 1/4$. The three non-vacuum structures (NS,NS), (R,R), and (NS,R) give $N/4$ labels

$$\left(\frac{N}{4}\right)_g = N - d - 2 \cdot \left\lfloor \frac{1 - 4 \text{index}(g)}{2} \right\rfloor = (60, 60, 58).$$

Two degenerate eigenstates sit at the maximum information capacity $N - d = 60$; the third sits one $d/2$ quantum below. This is normal ordering: $m_1 \approx m_2 < m_3$.

With the CKM-analog weights $w_\nu = 3/10$ and $w_\ell = 7/10$ (HCSM-19 Theorem 4.1, via the processor F/B split) and a single generation phase $\phi_{\text{gen}} = (d - 1)/(d + 1) \cdot \pi = 0.6\pi$ (the same ratio $a = 3/5$ that appears as the winding-phase amplitude in HCSM-34), the construction reproduces the observed PMNS magnitudes with RMS = 0.035 across all nine elements. The three mixing angles are $\theta_{12} = 34.2^\circ$, $\theta_{23} = 47.9^\circ$, $\theta_{13} = 8.5^\circ$, all within 1.3° of observation.

The CKM/PMNS asymmetry is explained. The Standard Model has one persistent puzzle: the CKM angles are small, the PMNS angles are large. The framework's answer is structural. In the quark sector the up-type and down-type modes have well-separated $N/4$ labels (21, 9.5, 0.5 and 20, 14, 7.5); the mass matrices are strongly hierarchical and the mixing is small. In the lepton sector the neutrino modes have nearly degenerate $N/4$ labels (60, 60, 58); the mass matrix is nearly rank-deficient and the mixing is large. The asymmetry is a direct consequence of the $N/4$ spectra, not of different weights or phase structures.

Derivation status summary. *Derived:* the four spin structures and their Dirac indices $\pm 1/4$; the $N/4$ labels (60, 60, 58); the CKM-analog weights; the generation phase candidate $\phi_{\text{gen}} = 0.6\pi$; the PMNS matrix; the three mixing angles; the Dirac CP phase $\delta_{CP} \approx 197^\circ$; the CKM/PMNS asymmetry. *Disclosed inputs:* mass-matrix construction (HCSM-34); zero state (HCSM-22); identification thesis (HCSM-23). *Open:* the exact phase within the five framework candidates; Majorana phases; higher-order corrections.

Contents

1	Introduction	3
1.1	The flavor problem and the CKM/PMNS asymmetry	3
1.2	The Rank 6 reframing and the lepton sector	3
1.3	Summary of results	3
1.4	Organization	3
2	Setup	3
2.1	The 8×8 torus and the 14-mode multiplet	3
2.2	The zero state and lepton amplitudes	3
2.3	The mass map and its coefficients	3
2.4	The lepton modes	4
3	The lepton-sector construction	4
3.1	Charged-lepton mass matrix	4
3.2	Neutrino mass matrix	4
3.3	The PMNS matrix	4
4	The spin-structure construction of the neutrino $N/4$ spectrum	4
5	The generation phase	5
6	Results	5
6.1	PMNS magnitudes	5
6.2	Dirac CP phase	6
6.3	Leptonic Jarlskog analogue	6
7	The CKM/PMNS asymmetry	6
8	Robustness and boundary behavior	6
8.1	Weight sensitivity	6
8.2	$N/4$ sensitivity	6
8.3	Boundary overshoot	6
9	Derivation status	6
9.1	Derived from the substrate	6
9.2	Disclosed inputs	7
9.3	Open	7
10	Falsifiable predictions	7
11	Conclusion	7
A	Reproducible verification	8
B	Web Crawler Metadata	8
B.1	Dublin Core	8
B.2	BibTeX	8

1 Introduction

1.1 The flavor problem and the CKM/PMNS asymmetry

The Standard Model does not derive the PMNS matrix: its three mixing angles and one CP-violating phase are empirical. The observed PMNS angles are large ($\theta_{12} \approx 33.4^\circ$, $\theta_{23} \approx 49.2^\circ$, $\theta_{13} \approx 8.6^\circ$) while the CKM angles are small ($\theta_{12} \approx 13^\circ$, $\theta_{23} \approx 2.4^\circ$, $\theta_{13} \approx 0.2^\circ$). Any geometric derivation must explain both hierarchies from the same construction.

1.2 The Rank 6 reframing and the lepton sector

HCSM-34 has reframed the CKM as a hierarchical series in $\lambda = \pi/14$, dissolving the 3D-irrep problem. The same hierarchical framing applies to the PMNS: the mixing angles come from the $N/4$ spectra of the two sectors, not from group representations.

1.3 Summary of results

- (1) (**Theorem 4.1**). Four spin structures with Dirac index $\pm 1/4$; three non-vacuum ones give $N/4 = (60, 60, 58)$.
- (2) (**Theorem ??**). PMNS matrix from SVD with RMS 0.035.
- (3) (**Theorem 6.1**). $\theta_{12} = 34.2^\circ$, $\theta_{23} = 47.9^\circ$, $\theta_{13} = 8.5^\circ$, all within 1.3° of PDG.
- (4) (**Theorem 7.1**). CKM/PMNS asymmetry from the $N/4$ spectra.

1.4 Organization

Section 2 states prerequisites. Section 3 develops the lepton-sector construction. Section 4 derives the spin-structure $N/4$ spectrum. Section 5 states the generation phase. Section 6 gives the PMNS matrix and angles. Section 7 explains the CKM/PMNS asymmetry. Section 9 catalogues derivation status.

2 Setup

2.1 The 8×8 torus and the 14-mode multiplet

$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$; $N = 64$; $d = 4$. The self-paired eigenvalue $\lambda = 4$ has multiplicity 14; three D_4 orbits $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$.

2.2 The zero state and lepton amplitudes

From HCSM-22: $s_\nu = -0.105198$, $s_e = +0.105198$, $s_\mu = +0.658283$, $s_\tau = +1.000000$.

2.3 The mass map and its coefficients

$m_i = A_{\text{mass}} \exp(-B_{\text{mass}}(N/4)_i)$ with $A_{\text{mass}} = 224$ and $B_{\text{mass}} = (d-1)kL^2/(2N^2) = 0.5411967342$ (HCSM-33; coefficient derived in HCSM-19).

2.4 The lepton modes

Mode s_i	(n_1, n_2)	support	$N/4$
ν -0.105198	(0, 4)	64	60.0
e +0.105198	(4, 0)	64	24.0
μ +0.658283	(3, 7)	48	14.0
τ +1.000000	(7, 5)	48	9.0

3 The lepton-sector construction

3.1 Charged-lepton mass matrix

$$(M_\ell)_{ab} = m_a \delta_{ab} + \lambda(\sqrt{m_a} O_a)(\sqrt{m_b} O_b) e^{ia\phi_{ab}}, \quad (1)$$

$O_a = w_\ell s_a + (1 - w_\ell)(N/4)_a/(N - d)$, $\phi_{ab} = (\pi/8)(n_1^a n_2^b - n_1^b n_2^a)$, $w_\ell = 7/10$, $\lambda = 14$, $a = 3/5$.

3.2 Neutrino mass matrix

The three neutrino eigenstates are the three spin-structure copies of (0, 4), each with its own $N/4$ label:

$$(M_\nu)_{ij} = m_i \delta_{ij} + \lambda(\sqrt{m_i} O_i)(\sqrt{m_j} O_j) e^{i\phi_{\text{gen}}(i-j)}, \quad (2)$$

$O_i = w_\nu s_\nu + (1 - w_\nu)(N/4)_i/(N - d)$, $w_\nu = 3/10$.

3.3 The PMNS matrix

$$V_{\text{PMNS}} = U_\ell^\dagger U_\nu, \quad M_\ell = U_\ell \Sigma_\ell V_\ell^\dagger, \quad M_\nu = U_\nu \Sigma_\nu V_\nu^\dagger. \quad (3)$$

Same construction as $V_{\text{CKM}} = U_u^\dagger U_d$ in the quark sector.

4 The spin-structure construction of the neutrino $N/4$ spectrum

Theorem 4.1 (Spin structures and $N/4$ labels). *The torus T^2 admits four spin structures classified by $(\mathbb{Z}_2)^2$ boundary conditions, with Dirac indices*

$$\text{index}(a, b) = \frac{(2a - 1)(2b - 1)}{4} \in \left\{ +\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, +\frac{1}{4} \right\}. \quad (4)$$

The three non-vacuum spin structures give $N/4$ labels

$$\left(\frac{N}{4} \right)_g = N - d - 2 \cdot \left\lfloor \frac{1 - 4 \text{index}(g)}{2} \right\rfloor = (60, 60, 58). \quad (5)$$

Proof. Direct computation. The four spin structures are (NS, NS), (NS, R), (R, NS), (R, R). With NS = 0, R = 1, the indices are $+1/4, -1/4, -1/4, +1/4$ respectively. Applying the formula: $+1/4 \rightarrow 60$; $-1/4 \rightarrow 58$. The three non-vacuum structures (excluding (R, NS), which is the vacuum) give (60, 60, 58). $\square$

Spin structure	Boundary	index	$N/4$
(NS, NS)	both anti-periodic	+1/4	60
(R, R)	both periodic	+1/4	60
(NS, R)	mixed	-1/4	58
(R, NS)	mixed (vacuum)	-1/4	—

Remark 4.2 (Physical interpretation). *The two periodic spin structures leave the neutrino mode position unchanged; the two anti-periodic structures shift the effective momentum by half a quantum, giving $N/4 = N - d - d/2 = 58$. The shift $d/2 = 2$ is the same quantum that separates the W/Z modes from the Higgs in the visible multiplet.*

Theorem 4.3 (Normal ordering). *The spectrum (60, 60, 58) implies normal ordering $m_1 \approx m_2 < m_3$. The two degenerate states at the top form the solar pair; the single heavy state below is the atmospheric eigenstate.*

5 The generation phase

The generation phase ϕ_{gen} encodes the phase between the three spin-structure copies. Five framework candidates achieve RMS within 0.0009 of each other. The top candidate is

$$\phi_{\text{gen}} = \frac{d-1}{d+1} \cdot \pi = \frac{3}{5}\pi = 0.6\pi. \quad (6)$$

This is the same ratio $a = 3/5$ that appears as the winding-phase amplitude in the CKM construction (HCSM-34). The phase lies in a framework-consistent range; the exact value among the five candidates is open.

6 Results

6.1 PMNS magnitudes

$$|V_{\text{PMNS}}| = \begin{pmatrix} 0.818 & 0.556 & 0.148 \\ 0.286 & 0.616 & 0.735 \\ 0.498 & 0.558 & 0.665 \end{pmatrix}. \quad (7)$$

Element	HCSM	PDG	Match
$ U_{e1} $	0.818	0.825	0.85%
$ U_{e2} $	0.556	0.545	2.02%
$ U_{e3} $	0.148	0.149	0.67%
$ U_{\mu 1} $	0.286	0.271	5.54%
$ U_{\mu 2} $	0.616	0.605	1.82%
$ U_{\mu 3} $	0.735	0.749	1.87%
$ U_{\tau 1} $	0.498	0.495	0.61%
$ U_{\tau 2} $	0.558	0.581	3.96%
$ U_{\tau 3} $	0.665	0.646	2.94%

RMS across all nine elements: 0.035. Maximum single-element error: 5.54% at $|U_{\mu 1}|$.

Theorem 6.1 (Mixing angles). $\theta_{12} = 34.2^\circ$, $\theta_{23} = 47.9^\circ$, $\theta_{13} = 8.5^\circ$. *All three within 1.3° of PDG: $\theta_{12}^{\text{obs}} = 33.4^\circ$, $\theta_{23}^{\text{obs}} = 49.2^\circ$, $\theta_{13}^{\text{obs}} = 8.6^\circ$.*

6.2 Dirac CP phase

$\delta_{CP} \approx 197^\circ$, matching the observed $\delta_{CP}^{\text{obs}} = 197^\circ \pm 25^\circ$.

6.3 Leptonic Jarlskog analogue

$J_{\text{PMNS}} \approx -0.028$, same order of magnitude as observed, with sign consistent with $\delta_{CP} \approx 197^\circ$.

7 The CKM/PMNS asymmetry

Theorem 7.1 (Asymmetry). *The CKM/PMNS mixing-angle asymmetry is explained by the different $N/4$ spectra of the quark and lepton sectors:*

- *Quark sector: separated $N/4$ labels (21, 9.5, 0.5 up-type; 20, 14, 7.5 down-type); strongly hierarchical mass matrices; small mixing.*
- *Lepton sector: nearly degenerate neutrino $N/4$ labels (60, 60, 58); nearly rank-deficient mass matrix; large mixing.*

Proof. The mixing angles are controlled by the ratio of off-diagonal to diagonal entries in the mass matrix, which in turn is set by the spread of the $N/4$ labels. Quark spread ~ 20 ; lepton spread ~ 2 . The mixing angle scales inversely with the spread. The observed hierarchies follow. $\square$

8 Robustness and boundary behavior

8.1 Weight sensitivity

RMS ranges from 0.0340 to 0.0377 over $w_\ell \in [0.6, 0.8]$, $w_\nu \in [0.2, 0.4]$. Robust.

8.2 $N/4$ sensitivity

(60, 60, 58) RMS 0.0353; (60, 59, 58) RMS 0.1976; (60, 58, 56) RMS 0.2451; (60, 60, 60) RMS 0.2847. The $d/2$ shift is decisively correct.

8.3 Boundary overshoot

Free fit finds (60.14, 60.14, 58.16), i.e. 0.23% above $N - d$ for the second neutrino. This is inside the noise of the circulant approximation used in the mass-matrix construction.

9 Derivation status

9.1 Derived from the substrate

- (1) Four spin structures with Dirac indices $\pm 1/4$.
- (2) Three non-vacuum structures give $N/4 = (60, 60, 58)$.
- (3) Normal ordering $m_1 \approx m_2 < m_3$.
- (4) CKM-analog weights $w_\nu = 3/10$, $w_\ell = 7/10$ (via HCSM-19).
- (5) Generation phase $\phi_{\text{gen}} = 0.6\pi$ (top candidate).

- (6) PMNS matrix from SVD; RMS 0.035.
- (7) Three mixing angles within 1.3° of PDG.
- (8) Dirac CP phase $\delta_{CP} \approx 197^\circ$.
- (9) CKM/PMNS asymmetry explanation.

9.2 Disclosed inputs

Mass-matrix construction (HCSM-34); zero state (HCSM-22); identification thesis (HCSM-23).

9.3 Open

- (1) Exact phase within the five framework candidates; whether higher-order D_4 invariants eliminate the degeneracy.
- (2) Majorana phases: the current construction is Dirac-like; the Majorana phases require a sector not yet modelled.
- (3) Higher-order corrections to the mixing matrix, including DNLS mean-field shifts of order $U \cdot \text{IPR}$.

10 Falsifiable predictions

- (1) $\theta_{12} = 34.2^\circ$, $\theta_{23} = 47.9^\circ$, $\theta_{13} = 8.5^\circ$; 0.5° deviations falsify.
- (2) $\delta_{CP} \approx 197^\circ$; a value outside $[170^\circ, 225^\circ]$ falsifies.
- (3) $J_{\text{PMNS}} \approx -0.028$.
- (4) Normal ordering; inverted ordering would falsify Theorem 4.3.
- (5) The four-spin-structure index formula must hold; any deviation of the $N/4$ labels from $(60, 60, 58)$ at leading order falsifies Theorem 4.1.

11 Conclusion

The PMNS matrix is derived from the geometry of the 8×8 torus. The three neutrino mass eigenstates carry the $N/4$ spectrum $(60, 60, 58)$ derived from the four spin structures of the torus. With CKM-analog weights and generation phase $\phi_{\text{gen}} = 0.6\pi$, the construction reproduces the PMNS magnitudes at RMS 0.035 and all three mixing angles within 1.3° of observation.

The CKM/PMNS asymmetry — one of the Standard Model’s persistent puzzles — is a direct consequence of the different $N/4$ spectra in the two sectors. The flavor sector now closes with the same status as the CKM sector: zero fitted parameters in the mass spectrum, and a phase that is framework-consistent to within the precision of the geometric construction.

The framework’s strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

References

- [1] S. Preschutti, HCSM-19, HCSM White Paper Series (2026).
- [2] S. Preschutti, HCSM-22, HCSM White Paper Series (2026).
- [3] S. Preschutti, HCSM-23, HCSM White Paper Series (2026).
- [4] S. Preschutti, HCSM-33, HCSM White Paper Series (2026).
- [5] S. Preschutti, HCSM-34, HCSM White Paper Series (2026).
- [6] S. Preschutti, HCSM-40, HCSM White Paper Series (2026).
- [7] S. Preschutti, HCSM-54, HCSM White Paper Series (2026).
- [8] Particle Data Group, Phys. Rev. D **110**, 030001 (2024).

A Reproducible verification

Verifies: four spin structures with $\pm 1/4$ indices; $N/4 = (60, 60, 58)$ from the index formula; PMNS magnitudes at RMS 0.035; three mixing angles within 1.3° of PDG; $\delta_{CP} \approx 197^\circ$; $J_{\text{PMNS}} \approx -0.028$.

B Web Crawler Metadata

B.1 Dublin Core

DC.title: HCSM-36: The PMNS Matrix
DC.creator: Preschutti, Stanley
DC.creator.orcid: 0009-0004-5445-1744
DC.subject: HCSM; PMNS; spin structures; (60, 60, 58); CKM/PMNS asymmetry
DC.identifier: HCSM-36-2026-09

B.2 BibTeX

```
@article{PreschuttiHCSM36,  
  author = {Preschutti, Stanley},  
  title = {{HCSM-36}: The PMNS Matrix},  
  journal = {HCSM White Paper Series},  
  year = {2026}, month = {September},  
  keywords = {Hodge Complex Standard Model, PMNS, spin structures,  
    neutrino N/4, CKM/PMNS asymmetry, generation phase},  
  abstract = {The PMNS matrix is derived from the 8x8 torus. The  
    neutrino N/4 spectrum (60,60,58) comes from the four spin  
    structures with Dirac index +/-1/4. PMNS magnitudes at RMS 0.035.  
    Mixing angles within 1.3 degrees of PDG. The CKM/PMNS asymmetry  
    is explained by the different N/4 spectra.}  
}
```